Work 2a - Your stack

Improve a shopping basket through acceptable product substitutions, nutritional constraints and a limited number of changes.

Component	Selected tool(s)	Level*	Used before?	Alternative
Input and output (steps 1, 11)	Streamlit + Pydantic + Plotly	2	Not confirmed	Gradio + dataclasses + Matplotlib
Data acquisition (step 2)	Open Food Facts on Hugging Face; fsspec + PyArrow	2	Not confirmed	OFF API + requests
Data preparation (step 3)	pandas + NumPy + RapidFuzz	2	Not confirmed	Polars + exact-match grouping
Pair annotation (step 4)	Label Studio + Python candidate rules	2	Not confirmed	Streamlit annotation form
Splits and baselines (steps 5, 6)	scikit-learn: GroupShuffleSplit, TF-IDF and cosine similarity	3	Not confirmed	Custom group split + category rules
AI: fine-tuning (step 7)	PyTorch + Sentence Transformers; MiniLM encoder + TripletLoss	3	Not confirmed	Frozen MiniLM encoder (no fine-tuning)
Retrieval and filtering (steps 8, 9)	NumPy cosine search + Python rules	2	Not confirmed	scikit-learn NearestNeighbors
Basket optimization (step 10)	OR-Tools CP-SAT	3	Not confirmed	Greedy substitution selection
Evaluation (step 12)	scikit-learn metrics + pytest; TensorBoard for training	3	Not confirmed	unittest + CSV metric logs
Reproducibility (step 13; ongoing)	Git + GitHub Actions; versioned data manifests	2	Not confirmed	GitLab + GitLab CI/CD

***Level:** provisional intermediate estimates (2-3), not verified against the course rubric. Confirm before submission.

Team experience: prior use has not been confirmed. Add member names or counts; do not interpret missing information as "no one".

Coaching priorities: check team experience with fine-tuning, leakage-safe pair construction and CP-SAT; request coaching where nobody has used the tools.

Project status: raw acquisition is implemented; the remaining stack is planned. The AI contribution is learned substitution ranking plus constrained optimization.